

Today, however, we have access to modern tools and approaches, and we no longer need to adhere to outdated methods.

India is known for its services, and I also ran a services company. Eventually, I recognised that controlling our future meant focusing on products, not services. Take SoCs (system-on-chip), for example: companies like Qualcomm attempted monolithic integration—putting everything into one giant chip. But that drove up cost, reduced yield, and customers disliked being locked into a single vendor.

Now the industry is shifting towards chiplets, which are modular components optimised for different processes and integrated at the package level. This approach allows greater flexibility, improved yields, and fosters innovation across various system elements.

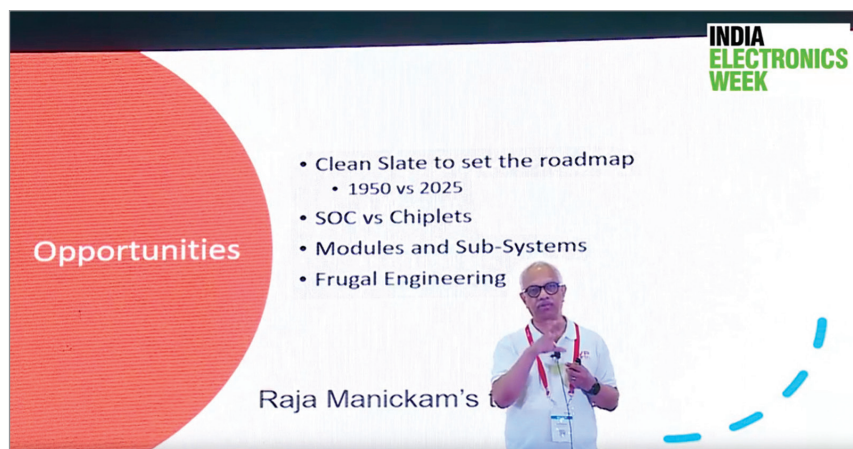
India, unencumbered by legacy baggage, is well positioned to adopt this model.

Frugal engineering and teamwork

We are naturally frugal, raised to make every rupee count. With today's exchange rates, our money stretches even further. But frugal engineering is not just about saving, it is about eliminating waste from the outset. That means designing products with efficient use of materials, resilient supply chains, and minimal resource loss. It is a mindset we grow up with.

One of India's biggest advantages is scalable human talent. Rather than everyone building separate teams, there is immense potential in pooling skilled resources across functions.

Imagine two companies—Company A and Company B—working in similar spaces. They may compete, but there is room for collaboration, particularly where one



Raja Manickam, during his engaging session at the Indian Electronics Week 2025, in Bengaluru

possesses expertise the other lacks. From design and marketing to prototyping, many skills are not available internally. If companies are willing to share even a few expert hours per week, they can reduce costs and amplify collective success.

At a macro level, two companies supporting each other grow stronger together. This is especially important for Indian startups, where building a chip requires diverse capabilities, from design and testing to packaging and business strategy. Unlike software, where most aspects are handled internally, hardware demands collaboration across firms and functions.

However, one of India's core challenges is a reluctance to collaborate. We are often raised to prioritise individual success. If someone else excels, the question becomes, "Why not you?" instead of "How can we succeed together?"

Yet chip development is not a one-person job. It requires multidisciplinary teamwork and the ability to recognise one another's strengths. In more collaborative cultures, children are taught early to work in teams and celebrate others' successes. That mindset, recognising that combining strengths yields better outcomes is essential for semiconductor innovation.

Frugal engineering also involves sharing resources. In manufacturing, we have been conditioned to believe that only mega fabs costing \$10 billion or more can succeed. These vast facilities, established by dominant players, create barriers for frugal innovators.

But it was not always this way. At TI, we had our own fab, built not to fill capacity but to support the product. The product drove the fab, not the other way around. That logic has since reversed. We must reconsider our approach if we are to foster innovation with limited resources.

The strategy of fabs

Fabs determine which products can be run and how customers interact with you. From a fab's perspective, the ideal scenario is continuous manufacturing of a single product. This minimises the need for innovation and process change. The preference is to follow a fixed process design kit (PDK), ensuring consistency and profitability.

Fabs seek large clients who can utilise full capacity, rather than smaller startups requesting limited wafer runs. They aim to maximise profit, often through high pricing, instead of pursuing lower-cost processes that depend on higher volume for similar revenue.